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$$A(z) = (z^2 - 1)(az + b)$$

$$A(0) = (0^2 - 1)(a0 + b)$$

$$A(0) = -b$$

$$A(0) = (2 + i)$$

$$-b = 2 + i$$

$$b = -2 - i$$

$$A(z) = (z^2 - 1)(az - 2 - i)$$

$$A(i) = 4$$

$$A(i) = (i^2 - 1)(ai - 2 - i)$$

$$A(i) = -2(ai - 2 - i)$$

$$A(i) = \frac{-2(ai - 2 - i)}{-2} = \frac{4}{-2}$$

$$A(i) = ai - 2 - i = -2$$

$$A(i) = ai - 2 + 2 - i = 0$$

$$A(i) = \frac{ai}{i} = \frac{i}{i}$$

$$a = 1$$

$$A(z) = (z^2 - 1)(z + (-2 - i))$$

$$A(z) = z^3 - (2 + i)z^2 - z + 2 + i$$

References